SEIF ALDEEN SAEED AHMED

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Mechatronics and Robotics Engineering student at Helwan National University with hands-on experience in autonomous robotics, embedded systems, power electronics, and Formula Student vehicle design. Vice Captain of Spark X Racing Team with practical experience in suspension system simulation, BLDC motor control, PCB prototyping, and robotics competitions. Passionate about building intelligent systems and solving real-world engineering challenges through both hardware and software integration.

EDUCATION

Helwan National University **2024 - 2028**

B.Sc. in Mechatronics and Robotics Engineering

- CGPA: 3.65 / 4.0 **(Ranked 3rd)**
- **Class Leader (2024 – 2026).**

EXPERIENCE

Intern **Aug – Sep 2025**

Citroën Service Center

- Learned mechanical maintenance procedures and automotive fault detection using Diagbox software.
- Practiced error clearing, module coding, and diagnostic workflows.
- Observed maintenance tasks such as AC evaporator replacement and timing belt installation.
- Gained exposure to suspension components and other maintainable vehicle parts.

Vice Captain – Autonomous Systems Member **Feb 2025 - Present**

Spark X Racing Team

- Contributed to the development of autonomous systems and robotics-related projects within the Formula Student team.
- Worked on embedded systems, control logic, and sensor integration for autonomous vehicle applications.
- Collaborated with technical and non-technical subteams during recruitment, workshops, and team activities.
- Supported team representation and coordination across different engineering disciplines.

Social Media Manager **Jul 2022 - Jul 2023**

H Technology – Computer Hardware Store

- Managed the brand’s social media presence, boosting engagement and visibility.
- Designed and scheduled promotional content tailored to the tech audience.
- Responded to customer inquiries and supported online communications.
- Improved branding consistency through creative visual content.

VOLUNTEERING

Organizer&Speaker – Orientation Day 2025 **Sep 2025**

Robotics & Mechatronics Engineering Dept. – Helwan National University

- Spoke to 350+ students about academic tips, learning resources, and competitions.
- Showcased personal projects **(Gesture Control Robot, Capture the Flag Robot).**
- Introduced **SparkX Racing Team** and emphasized the value of technical skills.

Speaker – LinkedIn Workshop **April 2025 & April 2026**

Robotics & Mechatronics Engineering Dept. – Helwan National University

- Delivered LinkedIn and personal branding workshops for Mechatronics & Robotics students.
- Guided students on building professional profiles, networking, and showcasing engineering projects effectively.
- Engaged 70+ students and encouraged them to improve their online professional presence.

COMPETITIONS

Auxilio Competition – 1st Place Winner
IEEE HSB | Capture the Flag Robotics Challenge

- Developed autonomous navigation and obstacle avoidance logic for a competitive maze-solving robot.
- Implemented control strategies and sensor integration for reliable real-time robot operation.
- Solved challenges related to motor synchronization, sensor calibration, and dynamic maze conditions.

RoboSoccer Competition – Participant

- Contributed to the development of an autonomous soccer robot with real-time movement and ball interaction capabilities.
- Worked on robot control strategies, navigation logic, and sensor integration.
- Improved practical understanding of autonomous robotics, debugging, and competition-based problem solving.

PROJECTS

- **BLDC Motor Controller (ESC)**

Designed and implemented a 3-phase BLDC motor controller using MOSFET inverter stages, IR2110 gate drivers, and PWM-based commutation control.

- **Formula Student Suspension System**

Designed a complete Formula Student suspension system using Autodesk Inventor, OptimumKinematics, and Simulink for dynamic analysis and suspension optimization.

- **Hand Gesture Control Car**

Built a wireless gesture-controlled robotic car using embedded systems, accelerometer sensors, and a custom-designed PCB glove controller.

- **Digital Clock Using Logic Gates**

Designed and implemented a fully functional digital clock using NE555 timers, 74LS90 counters, logic gates, and 7-segment displays.

- **Quick Return Mechanism (QRM) Shaping Machine Model**

Designed and built a mechanical model demonstrating the Quick Return Mechanism used in shaping machines for educational purposes.

TECHNICAL SKILLS

• **Robotics & Embedded Systems:** Arduino, ESP32, Embedded C/C++, Robotics Control, Autonomous Systems
• **CAD & Simulation:** Autodesk Inventor, MATLAB/Simulink, OptimumKinematics, Vehicle Dynamics
• **Electronics & Hardware:** PCB Design, Power Electronics, Sensors, Motor Drivers, Digital Logic
• **Automotive & Diagnostics:** Suspension Design, Mechanical Maintenance, Diagnostics (Diagbox)
• **Other:** Problem Solving, Team Leadership, Graphic Design, Social Media Management

SOFT SKILLS

- **Leadership & Team Management**
- **Problem Solving & Critical Thinking**
- **Communication & Public Speaking**
- **Team Collaboration**
- **Project Coordination**

LANGUAGES

- **Arabic: Native**
- **English: Fluent (C1)**